

WJEC GCSE

Digital Technology

Approved by Qualifications Wales

Non-Examination Assessment Handbook

Unit 2: Digital Practices (3490U2)

Teaching from 2026

For award from 2028



Contents

About this pack: information for teachers	1
Managing the assessment	1
About this pack: information and instructions for candidates.....	6
Appendix A: Digital Skills.....	32
Appendix B: Mapping of questions to specification content and assessment objectives	36

Copyright

© WJEC CBAC Limited 2026.

About this pack: information for teachers

This pack contains the assessment materials for Unit 2: Digital Practices.

Within this pack, you will find:

- information for teachers on how to manage arrangements for assessment
- overview of candidate requirements
- research requirements
- assessment information: a detailed breakdown of each task, the evidence and any additional instructions related to the task, the assessment conditions (controls) under which the task is to be completed, timing and resources
- appendix of skills
- the mark scheme which must be used to assess candidate work
- information for candidates, which tells candidates about things that they must and must not do when they are completing the assessment.

You must provide each candidate with a copy of the candidate assessment pack and the client brief for the relevant NEA submission year.

Managing the assessment

This is a centre-marked (internal) non-examination assessment. The following arrangements must be followed when managing and marking the assessment.

Task setting

This assignment has been produced by WJEC and cannot be changed. The client brief will be issued annually by WJEC and made available on the WJEC Portal in September.

Assessment timeframe

Assessment may be undertaken at any time during the qualification. However, centres should ensure that assessment is completed only when learners have undertaken the necessary teaching and learning and developed the required skills and knowledge.

Time

Candidates must be allowed **20 hours** to complete this assessment. Candidates should spend approximately:

- 2.5 hours completing task 1
- 1 hour completing task 2
- 2.5 hours completing task 3
- 13 hours completing task 4
- 1 hour completing task 5

The time suggested for each task takes account of the contribution of the task to the overall assessment requirements. There can be no changes to the total time available for this assessment. Centres can, however, amend the suggested time available for each task.

Research

Candidates can use research collected in non-supervised sessions.

Referencing

Candidates must reference any research brought into/developed during the NEA. Candidates must follow the guidelines on 'When to Reference a Source' found in Appendix 1 and 'How to Reference a Source' found in Appendix 2 of the document named 'Conducting Non-examination Assessments in National 14-16 Qualifications – Guide for Teachers'.

Assessment conditions

There are three levels of Assessment conditions: high, medium and low. The conditions relevant for this unit are stated below.

Level	Assessment conditions
High	Candidates complete work under formal supervision¹ – in a classroom, (e.g. with teacher supervision) following these rules: • No communication with peers is allowed (including unrelated topics) • No assistance may be given by the teacher beyond reading the instructions • Candidates must work independently, and all work must be their own • Candidate's work must always remain within the centre and stored securely between timetabled sessions • Wall displays or materials that could advantage candidates must not be visible • In line with JCQ regulations for non-examination assessments, the 'Warning to Candidates' and 'Unauthorised Items Poster' must be displayed.

¹ There are no set requirements regarding the number of teachers/assessors required to be present during an assessment. However, the ratio must be sufficient to ensure appropriate supervision of candidates throughout. If only one teacher/assessor is present, they must be able to request assistance without needing to leave the assessment room. Under no circumstances should candidates be left unsupervised or unattended at any time.

Resources

There are three levels of resource conditions: none, specified and permitted. The relevant conditions for this unit are stated below.

Level	Resources
Specified	• Candidates may access only the specific resources stated in the Research Information and/or Assessment Information. This may include centre-provided resources or research candidates collected during the research phase • Internet access (and other resources) may be allowed for research if specified by the Assessment Information; teachers must monitor this • Candidates must not provide resources unless explicitly stated in the Assessment Information • Candidates must follow the 'When to Reference a Source' found in Appendix 1 and 'How to Reference a Source' found in Appendix 2 when using resources found in 'Conducting Non-examination Assessments in National 14-16 Qualifications – Guide for Candidates'

Collaboration

Group work is **not** allowed. Candidates must complete all tasks individually

Supervision

Candidates must be supervised by an assessor whilst completing the activities. Centres must have in place systems to ensure candidates cannot access evidence they have been developing outside of supervised activities.

Authentication

Supervision must be in place to ensure the authenticity of evidence produced for summative assessment.

You should not provide input or guidance to candidates during the assessment time. This includes providing formative feedback on the evidence being produced. You can provide guidance on the requirements of the task and remind candidates of the mark bands and how they can be interpreted.

Candidates can review and redraft evidence independently within the time controls for the assessment.

Candidates must sign the declaration to confirm that all evidence submitted for assessment is their own work and that any sources, including any AI, used have been acknowledged.

Assessors must sign the declaration to confirm that evidence submitted for assessment was completed under the controlled conditions set out in the assessment.

Candidate evidence

The acceptable formats for evidence of each of the tasks is summarised in the task tables. Any modified assessment must enable the candidate to generate the evidence required in this table.

Marking the assessment

The marking scheme for this assessment is included in this handbook.

All marking of evidence must be made against the marking scheme provided. Evidence marked must comply with any requirements set out in this handbook.

All provided evidence must be annotated to show how it relates to the mark band requirements.

Marking should only be undertaken by a designated assessor. An assessor should have appropriate expertise in the subject and level for a specified unit. The assessor is responsible for ensuring that:

- assessment is conducted under the controls specified in the task table
- they are clear about the requirements of the assessment and marking scheme prior to commencing assessment
- evidence presented for assessment is valid, authentic; reliable and credible
- assessment decisions are accurately recorded
- evidence is appropriately annotated
- judgements are only made against the mark band statements.

Centres must ensure that a candidate mark record sheet is completed for every candidate and is included with each candidate's work.

Centres must also ensure that candidates complete a candidate declaration form.

Guidance for Assessment

Each task has a separate assessment grid.

The appropriate band should be established by determining which performance descriptor best reflects the learner's performance.

The appropriate mark should then be established by determining the extent to which the learner has met the criteria within the relevant band – completely met (upper part of band), mostly met (middle of band), partially met (lower part of band).

Notes on the completion of the Candidate Declaration Form

- The form must be completed and included with candidate's work, and uploaded to WJEC along with the work.
- The form can be completed electronically or manually. Electronic signatures will be accepted.

Further information

You should refer to the JCQ Instructions for conducting non-examination assessment for further information on managing assessment within our GCSE Qualifications.

About this pack: information and instructions for candidates

Handbook overview

This information is for candidates completing GCSE Digital Technology Unit 2: Digital Practices Non-Examination Assessment.

Within this pack, you will find:

- overview of candidate requirements
- information for candidates, which tells you about things that you must and must not do when you are completing the assessment
- research information
- assessment information: a detailed breakdown of each task, the evidence requirements and the conditions (controls) under which the task is to be completed
- appendix of skills that should be used when developing your project
- the mark scheme which will be used to assess your work.

Assessment management

Time

You will have 20 hours in total to complete this assessment. You should spend approximately:

- 2.5 hours completing task 1
- 1 hour completing task 2
- 2.5 hours completing task 3
- 13 hours completing task 4
- 1 hour completing task 5.

Supervision

You will be supervised throughout the assessment.

You do not need to be supervised when collecting digital assets. This collection of resources does not form part of your non-examination assessment time.

Monitoring

- Teachers may provide guidance and support to you to ensure that you have a clear understanding of the requirements, including the brief and the assessment grid.
- Once the creating and developing work is underway, feedback must be limited to general advice on what needs to be improved. Teachers must not provide specific guidance on how to achieve these improvements.
- Teachers must see your ideas in progress at different stages during the process to be able to authenticate the work as your own and complete the necessary authentication documentation.

It is important that you work independently from other candidates and the work produced should be your own unaided work.

By submitting your work and signing the candidate declaration form, you confirm that your work complies with all assessment controls and guidance, including the rules on plagiarism and the responsible use of Artificial Intelligence (AI). All work must be your own, produced independently and without unauthorised assistance. Both you and your teacher will be required to sign a declaration confirming that the work submitted is entirely your own.

You are advised to check your work carefully to make sure that it is accurate and correct.

Resources

You are permitted to collect digital assets to use within your digital products. These are collected outside your non-examination assessment time.

Assessment Details

For this assessment you are required to complete a series of tasks which relate to a given client brief. You are required to use and organise given datasets purposefully. You are also required plan and develop data-informed digital products, of either a computer game or a 2D animation, for a specific audience and purpose. Finally, you are required to test and evaluate the success of your created digital product.

This assessment contributes to 40% of the overall qualification grade and will take 20 hours to complete. The assignment will be marked out of a total of 80 marks.

Based on the client brief, you are required to collect digital assets to prepare for the non-examination assessment.

Overview of requirements

You are required to:

Outside non-examination assessment time:

- collect digital assets.

During the non-examination assessment time:

In Task 1, you are required to:

- cleanse and organise data to ensure it is suitable for data analytics
- use data analytics to identify audience needs, trends and preferences.

In Task 2, you are required to:

- create a digital product proposal
- create a time planning document.

In Task 3, you are required to:

- plan annotated digital design options for a digital product
- justify your choice of design options.

In Task 4, you are required to:

- design, create and test two functioning digital products including a multi-page website and either a computer game or a 2D animation.

In Task 5, you are required to:

- evaluate your approach to testing and the functionality of your digital products.

Digital asset collection

To prepare for the non-examination assessment, you should collect digital assets to include within your digital products. Collecting digital assets does not form part of the non-examination assessment time. The collection of digital assets does not need to take place all at once – you are permitted to gather and add to your digital assets at any point during the assessment window. Collecting digital assets does not form part of your assessed tasks, but you will need these to allow you to complete your assessed tasks.

Artificial Intelligence (AI)

The following information is taken from the [JCQ Information for candidates – non-examination assessments](#) and the [JCQ Artificial Intelligence \(AI\) Use in Assessments: Protecting the Integrity of Qualifications](#). This tells you about things that you **must** and **must not** do when you are completing your assessment. If there is anything that you do not understand, you **must** ask your teacher.

Plagiarism

Plagiarism involves taking someone else's words, thoughts or ideas and trying to pass them off as your own. **It is a form of cheating which is taken very seriously.** There are many ways to detect plagiarism:

- Markers can spot changes in the style of writing and use of language.
- Markers are highly experienced subject specialists – they may have read the source you are using or even marked the work you have copied from!
- Internet search engines and specialised computer software can be used to match phrases or pieces of text with original sources and to detect changes in the grammar and style of writing or punctuation.

Penalties for breaking the regulations

If it is discovered that you have broken the regulations, you will be referred to the Malpractice panel who might impose a penalty.

WJEC will decide which penalty is appropriate.

Remember – it's your qualification so it needs to be your own work

Tasks

	Task	Evidence	Controls
1	Use the quantitative dataset provided: Perform the following tasks using suitable spreadsheet skills, formulas and functions. (a) Cleanse and organise the dataset to enable data analytics to take place. You should use a range of suitable spreadsheet skills categorised as basic, medium and complex (as listed in the spreadsheet skills, Appendix A). [5 marks]	Cleansed and organised dataset. Screen shot with concise written annotation or oral comments in a screen recording with concise explanations (written or oral) where spreadsheet skills are not obvious to the assessor.	Assessment conditions: High Resources: Specified
	(b) Using your cleansed and organised dataset, apply suitable data analytic skills to identify: • audience needs • audience trends • audience preferences. Present your findings in a suitable format - which should include appropriate graphs or charts. [5 marks]	Findings presented in a suitable format.	
Permitted resources: Dataset file as specified in your client brief			

	Task	Evidence	Controls
2	Develop a digital project proposal for the multipage website and chosen digital product. You should use the client brief, audience needs, trends and preferences, and the qualitative data provided. (a) Your digital project proposal must include: Specific and measurable success criteria that includes: • alignment with data analytics results • design aspects that reflect qualitative data provided • functionality and usability. [5 marks] (b) Time planning document to ensure the development process is completed on time, including: • tasks • activities • milestones. [5 marks]	Written document Time planning document. Please note a multipage website consists of 3 pages.	Assessment conditions: High Resources: Specified
Permitted resources: Qualitative data as specified in the client brief			

	Task	Evidence	Controls
3	Using your success criteria, client brief and qualitative data, complete the following tasks: (a) Plan annotated initial design options for: • a multipage website • your chosen digital product. [10 marks] Please note. You must plan at least two design options for each digital product.	Two initial design options for a multipage website. Two initial design options for your chosen digital product. Please note a multipage website consists of three pages.	Assessment conditions: High Resources: Specified
	Select the chosen design option that you will implement for your multipage website and chosen digital product. Complete the following task based on your selection: (b) Justify your choices of selected design options based on your success criteria. 5 marks]	A written justification	
Permitted resources: Qualitative data as specified in the client brief Digital assets			

	Task	Evidence	Controls
4	(a) Create a functional multipage website which uses the information provided by the client and which reflects your chosen design option from Task 3. Your website should use: • suitable media • effective navigation between pages • hyperlinks to external documents • consistent layout and styles. You should use a range of suitable website development skills categorised as basic, medium and complex (as listed in Appendix A) when creating your multipage website. [10 marks]	Screen shot evidence with concise written annotation or oral comments in a screen recording with concise explanations (written or oral) where website development skills are not obvious to the assessor. Conduct ongoing testing to ensure your multipage website and digital product meets your success criteria and functions as expected. No evidence of this testing is required. Exported website in hostable format (HTML, CSS & JS files) Please note a multipage website consists of 3 pages.	Assessment conditions: High Resources: Specified

	(b) Using your selected design option, create a multilevel computer game that uses: • scripting-based languages to control game behaviour • game characters, objects and environments including animation • sound effects and background music to enhance the gaming experience • intuitive, interactive and engaging interface. You should use a range of suitable game development techniques from each level of complexity (categorised as basic, medium and complex, as listed in Appendix A). [25 marks] Or (b) Using your selected design options, create a 2D animation that uses: • suitable animation production skills • suitable sound to enhance the animation • multiple animation sets to depict various scenes and actions • animation timeline skills to control the timing and movement of animations. You should use a suitable range of animation techniques from each level of	Screen shot with concise written annotation or oral comments in a screen recording with concise explanations (written or oral) where computer game development skills or animation skills are not obvious to the assessor. Exported computer game as a standalone application downloadable from your website. <u>OR</u> Exported animation embedded / playable into the website. Conduct ongoing testing to ensure your multipage website and digital product meets your success criteria and functions as expected. No evidence of this testing is required.	
--	---	---	--

	complexity (categorised as basic, medium and complex, as listed in Appendix A). [25 marks]		
Permitted resources Qualitative data as specified in the client brief Digital assets Chosen design options			

	Task	Evidence	Controls
5	(a) Evaluate the effectiveness of your ongoing testing of your digital products. Refer to how your testing helped you to: • identify and remove errors • ensure the digital products operated as expected • meet the success criteria. [5 marks]	Written document No screen shot evidence of testing is required	Assessment conditions: High Resources: Specified
	(b) Evaluate how well your digital products meet your audience needs, trends and preferences. Refer to your: • completed digital products • development process • application of skills. [5 marks]	Written document No screen shot evidence of testing is required	
No new resources apart from resources specified in Tasks 1, 2, 3 and 4.			

End of tasks

Mark Scheme

Guidance for assessors

Assessment grids for non-examination assessment marking principles

Banded mark schemes

Banded mark schemes are divided so that each band within a section has a relevant descriptor. The descriptor for the band provides a description of the performance level for that band. Each band contains marks. Before marking, the teacher should first read and annotate a candidate's project to pick out the evidence that is being assessed. Once the annotation is complete, the mark scheme can be applied.

This is done as a two-stage process:

Stage 1 – Deciding on the band

Beginning at the lowest band, the teacher should look at the appropriate section of the candidate's work and check whether it matches the descriptor for that section's mark band. The teacher should look at the descriptor for that band and see if it matches the qualities shown in the candidate's work for that section. If the descriptor at the lowest band is satisfied, the teacher should move up to the next band and repeat this process for each band until the descriptor matches the work. If a candidate's work covers different aspects of different bands within the mark scheme, a 'best fit' approach should be adopted to decide on the band and then the candidate's work should be used to decide on the mark within the band. For instance, if work is mainly in band 2 but with a limited amount of band 3 content, the work would be placed in band 2, but the mark awarded would be close to the top of band 2 as a result of the band 3 content. Teachers should not seek to mark candidates down as a result of small omissions in minor areas of their work.

Stage 2 – Deciding on the mark

Once the band has been decided, the teachers can then assign a mark. The teachers are reminded of the need to revisit the work as they apply the mark scheme in order to confirm that the band and the mark allocated is appropriate to the work submitted. Where work is not credit worthy, that is, contains nothing of any significance to the project, or has been omitted, no marks should be awarded. Internal standardisation is essential, that where there is more than one teacher in a centre, work from all teaching groups is standardised internally. This is designed to ensure that the final assessment reflects a single agreed standard for all teaching groups involved

Unit 2 Mark scheme

The following mark scheme will be used to assess your work.

0 marks will be awarded if there is no evidence or submitted work is not worthy of credit.

Task 1

Use the quantitative dataset provided:

Perform the following tasks using suitable spreadsheet skills, formulas and functions.

(a) Cleanse and organise the dataset to enable data analytics to take place.

You should use a range of suitable spreadsheet skills categorised as basic, medium and complex (as listed in the spreadsheet skills, Appendix A).

Band	AO3: Apply knowledge, skills and understanding to identify audience needs and plan digital products and communications that meet these needs.
3	5 marks - comprehensive application of a range of suitable spreadsheet skills, formulas and functions to produce fully cleansed and organised data - the response includes the use of at least three complex and three other (either basic or medium) suitable spreadsheet skills as categorised in Appendix A.
2	3-4 marks - adequate application of suitable spreadsheet skills, formulas and functions to produce adequately cleansed and organised data. - the response includes the use of at least two complex and two other (either basic or medium) suitable spreadsheet skills as categorised in Appendix A.
1	1-2 marks - basic application of suitable spreadsheet skills, formulas and functions to attempt to cleanse and organise data. - the response includes the use of at least one complex and one other (either basic or medium) suitable spreadsheet skill as categorised in Appendix A.

0	0 marks No response attempted or nothing worthy of credit.
----------	--

Task 1

(b) Using your cleansed and organised dataset, apply suitable data analytic skills to identify:

- audience needs
- audience trends
- audience preferences.

Present your findings in a suitable format - which should include appropriate graphs or charts

Band	AO3: Apply knowledge, skills and understanding to identify audience needs and plan digital products and communications that meet these needs.
3	5 marks • comprehensive application of knowledge, skills and understanding of data analytics to identify audience needs, trends and preferences • the findings are presented in a very clear and well-structured format which includes appropriate graphs or charts
2	3-4 marks • satisfactory application of knowledge, skills and understanding of data analytics to identify audience needs, trends and preferences • the findings are presented in a clear and structured format which includes mostly appropriate graphs or charts
1	1-2 marks • basic application of knowledge, skills and understanding of data analytics to identify either audience needs, trends or preferences • the findings are presented in an unclear or unstructured format, graphs or charts, and may not be appropriate
0	0 marks No response attempted or nothing worthy of credit.

Task 2

Develop a digital project proposal for the multipage website and chosen digital product. You should use the client brief, audience needs, trends and preferences, and the qualitative data provided.

(a) Your digital project proposal must include:

Specific and measurable success criteria that includes:

- alignment with data analytics results
- design aspects that reflect qualitative data provided
- functionality and usability.

Band	AO3: Apply knowledge, skills and understanding to identify audience needs and plan digital products and communications that meet these needs.
3	5 marks • comprehensive application of knowledge and understanding of audience needs, trends and preferences to develop specific and measurable success criteria that includes all of the following: • alignment with data analytics results • design aspects that reflect qualitative data provided • functionality and usability • the success criteria clearly links relevant aspects of the client brief and the qualitative data
2	3-4 marks • adequate application of knowledge and understanding of audience needs, trends and preferences to develop specific and measurable success criteria that includes some of the following: • alignment with data analytics results • design aspects that reflect qualitative data provided • functionality and usability • the success criteria links relevant aspects of the client brief and the qualitative data

1	1-2 marks • basic application of knowledge and understanding of either audience needs or trends or preferences to develop specific or measurable success criteria that includes at least one of the following: • alignment with data analytics results • design aspects that reflect qualitative data provided • functionality and usability • the success criteria makes minimal links between relevant aspects of the client brief or qualitative data
0	0 marks No response attempted or nothing worthy of credit.

Task 2

(b) Time planning document to ensure the development process is completed on time, including:

- tasks
- activities
- milestones.

Band	AO3: Apply knowledge, skills and understanding to identify audience needs and plan digital products and communications that meet these needs.
3	5 marks • comprehensive application of knowledge and understanding of time planning to ensure the digital products are completed on time • the time planning document effectively and correctly communicates all tasks, activities and milestones
2	3-4 marks • adequate application of knowledge and understanding of time planning to ensure the digital products are completed on time • the time planning document communicates some tasks, activities and milestones
1	1-2 marks • basic application of knowledge and understanding of time planning • the time planning document communicates minimal tasks or activities or milestones

0	0 marks No response attempted or nothing worthy of credit.
----------	--

Task 3

Using your success criteria, client brief and qualitative data, complete the following tasks:

(a) Plan annotated initial design options for:

- a multipage website
- your chosen digital product.

Please note. You must plan at least two design options for each digital product.

Band	AO3: Apply knowledge, skills and understanding to identify audience needs and plan digital products and communications that meet these needs.
5	9-10 marks • the annotated initial design options demonstrate excellent application of knowledge and understanding of effective design for all digital products • annotations communicate all success criteria, all relevant aspects of the client brief and all relevant qualitative data • all sound aspects and all visual aspects are communicated in the initial design options.
4	7-8 marks • the annotated initial design options demonstrate very good application of knowledge and understanding of effective design for all digital products • annotations communicate most success criteria, most relevant aspects of the client brief and most relevant qualitative data • most sound aspects and most visual aspects are communicated in the initial design options.
3	5-6 marks • the annotated initial design options demonstrate good application of knowledge and understanding of effective design for most digital products • annotations communicate some success criteria, some relevant aspects of the client brief and some relevant qualitative data • some sound aspects and some visual aspects are communicated in the initial design options.

2	3-4 marks - the annotated initial design options demonstrate basic application of knowledge and understanding of effective design for most digital products - annotations communicate minimal success criteria, minimal aspects of the client brief and minimal qualitative data - minimal sound aspects and minimal visual aspects are communicated in the initial design options.
1	1-2 marks - the annotated initial design options demonstrate limited application of knowledge and understanding of effective design for some digital products - annotations communicate minimal success criteria or minimal aspects of the client brief or minimal qualitative data - minimal sound aspects or minimal visual aspects are communicated in the initial design options.
0	0 marks No response attempted or nothing worthy of credit.

Task 3

Select the chosen design option that you will implement for your multipage website and chosen digital product.

Complete the following task based on your selection:

(b) Justify your choices of selected design options based on your success criteria.

Band	AO4: Develop, test and evaluate digital products that meet the needs of specified audiences.
3	5 marks - comprehensive justification of the selected design option for the multipage website that reflects all aspects of the success criteria - comprehensive justification of the selected design option for the chosen digital product that reflects all aspects of the success criteria.
2	3-4 marks - adequate justification of the selected design option for the multipage website that reflects most aspects of the success criteria - adequate justification of the selected design option for the chosen digital product that reflects most aspects of the success criteria.

1	1-2 marks • basic justification of the selected design option for the multipage website that reflects some aspects of the success criteria • basic justification of the selected design option for the chosen digital product that reflects some aspects of the success criteria.
0	0 marks No response attempted or nothing worthy of credit.

Task 4

(a) Create a functional multipage website which uses the information provided by the client and which reflects your chosen design option from Task 3. Your website should use:

- suitable media
- navigation
- hyperlinks
- consistent layout and styles.

You should use a range of suitable website development skills categorised as basic, medium and complex (as listed in Appendix A) when creating your multipage website.

Band	AO4: Develop, test and evaluate digital products that meet the needs of specified audiences.
5	9-10 marks • the multipage website is fully functional and includes all relevant information provided in the client. It contains all the following: • suitable media • effective navigation between pages • hyperlinks to external documents • consistent layout and styles • a well-developed multipage website that reflects all aspects of the chosen design option • at least five complex, five medium and five basic relevant website development skills are used effectively in the production of the website.

4	7-8 marks - the multipage website is functional and includes most relevant information provided by the client. It contains most of the following: - suitable media - effective navigation between pages - hyperlinks to external documents - consistent layout and styles - a developed multipage website that reflects most aspects of the chosen design option - at least four complex, four medium and four basic relevant website development skills are used in the production of the website.
3	5-6 marks - the multipage website is partially functional and includes some relevant information provided by the client, as well as some of the following: - suitable media - effective navigation between pages - hyperlinks to external documents - consistent layout and styles - an adequately developed multipage website that reflects some aspects of the chosen design option - at least three complex, three medium and three basic relevant website development skills are used to produce the website.
2	3-4 marks - the multipage website is basic with minimal functionality, it includes information provided by the client with minimal relevance. It includes at least two of the following list: - suitable media - effective navigation between pages - hyperlinks to external documents - consistent layout and styles - a simple multipage website that reflects minimal aspects of the chosen design option - at least six complex, medium or basic website development skills are demonstrated.

1	1-2 marks - the multipage website is limited and not functional, information is mostly irrelevant. It contains at least one of the following list: - suitable media - effective navigation between pages - hyperlinks to external documents - consistent layout and styles - a poorly developed multipage website that barely reflects the chosen design option - at least three complex, medium or basic website development skills are demonstrated.
0	0 marks No response attempted or nothing worthy of credit.

Task 4

(b) Using your selected design option, create a multilevel computer game that uses:

- scripting-based languages to control game behaviour
- game characters, objects and environments including animation
- sound effects and background music to enhance the gaming experience
- intuitive, interactive and engaging interface.

You should use a suitable range of suitable game development techniques from each level of complexity (categorised as basic, medium and complex, as listed in Appendix A)

Band	AO4: Develop, test and evaluate digital products that meet the needs of specified audiences.
5	21-25 marks • the computer game demonstrates excellent development of all technical skills including: • highly effective control of game behaviour • advanced use of designing and using game characters, objects and environments • the computer game is multilevel • includes sound effects and background music which are highly effective in enhancing the gaming experience • all interface elements are intuitive, interactive and engaging • at least five complex, five medium, and five basic game development techniques are applied, which are all highly suitable • a highly-developed computer game that fully reflects all success criteria, chosen design options and qualitative data.
4	16-20 marks • the computer game demonstrates very good development of most technical skills, including: • effective control of game behaviour • good use of designing and using game characters, objects, and environments • the computer game is multilevel • includes sound effects and background music, which enhance the gaming experience • most interface elements are intuitive, interactive and engaging • at least four complex, four medium, and four basic game development techniques are applied, which are nearly all suitable • a developed computer game that reflects most success criteria, chosen design options and qualitative data.

3	11-15 marks - the computer game demonstrates adequate development of technical skills, including: - some control of game behaviour - some designing and using game characters, objects and environments - the computer game has multiple levels, though some may lack depth - includes some sound effects and background music, but their effectiveness is limited - some interface elements are interactive and engaging, but there may be inconsistencies - at least three complex, three medium, and three basic game development techniques are applied, which are somewhat suitable - a functional computer game that reflects some success criteria, chosen design options, and qualitative data.
2	6-10 marks - the computer game demonstrates basic development of technical skills, including: - minimal control of game behaviour - minimal game characters, objects, and environments - the computer game may lack multiple levels or have poorly developed levels - includes few sound effects or background music, which may not enhance the experience - some interface elements are present but lack interactivity or engagement - at least six complex, medium or basic game development techniques is applied, which are somewhat suitable - a partially developed computer game that reflects few success criteria, chosen design options, and qualitative data.
1	1-5 marks - the computer game demonstrates limited development of technical skills, including: - limited or missing control of game behaviour - limited or missing game characters, objects, and environments - the computer game is single-level or incomplete - limited or no sound effects or background music - interface elements are present but lack functionality or engagement - at least three complex, medium, or basic game development techniques are applied, which lack suitability - an undeveloped computer game that reflects very few success criteria, chosen design options, and qualitative data.
0	0 marks No response attempted or nothing worthy of credit.

Task 4

Using your selected design options, create a 2D animation that uses:

- suitable animation production skills
- suitable sound to enhance the animation
- multiple animation sets to depict various scenes and actions
- animation timeline skills to control the timing and movement of animations.

You should use a suitable range of animation techniques from each level of complexity (categorised as basic, medium and complex, as listed in Appendix A).

Band	AO4: Develop, test and evaluate digital products that meet the needs of specified audiences.
5	21-25 marks • the animation demonstrates excellent development of all technical skills, including: • highly effective design of a range of characters, including animated sprites • advanced use of suitable animation production skills • embedded sound is well-synchronised and enhances the animation effectively • multiple animation sets are used effectively to depict various scenes and actions • precise control of timing and movement is evident through skilful use of the animation timeline • at least five complex, five medium and five basic skills animation techniques are applied, which are all highly suitable • a highly developed animation that fully reflects all success criteria, chosen design options and qualitative data.
4	16-20 marks • the animation demonstrates very good development of most technical skills, including: • effective design of a range of characters, including animated sprites • good use of suitable animation production skills • embedded sound enhances the animation, though minor synchronisation issues may be present • multiple animation sets are used, though some transitions may lack refinement • good control of timing and movement through the animation timeline • at least four complex, four medium and four basic skills animation techniques are applied, which are nearly all suitable • a developed animation that reflects most success criteria, chosen design options and qualitative data.

3	11-15 marks - the animation demonstrates adequate development of technical skills, including: - some variety in character design, including animated sprites - some use of suitable animation production skills - embedded sound is included but may not always align with the animation effectively - multiple animation sets are present but may lack consistency in movement or transitions - some control of timing and movement, but inconsistencies are evident - at least three complex, three medium and three basic skills animation techniques are applied, which are somewhat suitable - a functional animation that reflects some success criteria, chosen design options and qualitative data.
2	6-10 marks - the animation demonstrates basic development of technical skills, including: - limited variety in character design, with some animated sprites - minimal use of animation production skills, focusing mainly on basic techniques - embedded sound may be included but is poorly synchronised or ineffective - few animation sets are used, with noticeable gaps in continuity - basic control of timing and movement, with evident inconsistencies - at least six complex, medium or basic skills animation techniques is applied, which are somewhat suitable - a partially developed animation that reflects few success criteria, chosen design options and qualitative data.
1	1-5 marks - the animation demonstrates limited development of technical skills, including: - limited variety in character design, with few or no animated sprites - limited or missing animation production skills, relying only on basic movement with no refinement - no embedded sound or sound is completely misaligned with the animation - limited animation sets, with little effort to depict different scenes or action. - limited control of timing and movement, leading to a disjointed animation - at least three complex, medium, or basic skills animation techniques are applied, which lack suitability - an undeveloped animation that reflects very few success criteria, chosen design options and qualitative data.
0	0 marks No response attempted or nothing worthy of credit.

Task 5 (a)

Evaluate the effectiveness of your ongoing testing of your digital products. Refer to how your testing helped you to:

- identify and remove errors
- ensure the digital products operated as expected
- meet the success criteria.

Band	AO4: Develop, test and evaluate digital products that meet the needs of specified audiences.
3	5 marks • the response is comprehensive, demonstrating a detailed evaluation of how ongoing testing helped achieve the following: • identify and remove errors • ensure the digital products operated as expected • meet the success criteria • a comprehensive evaluation of the impact of testing at different stages
2	3-4 marks • the response is adequate, demonstrating a sound evaluation of how ongoing testing helped achieve the following: • identify and remove errors • ensure the digital products operated as expected • meet the success criteria • an adequate evaluation of the impact of testing at different stages, though coverage may lack some depth or detail.
1	1-2 marks • The response is basic, demonstrating limited evaluation of how ongoing testing helped achieve the following: • identify and remove errors • ensure the digital products operated as expected • meet the success criteria • a basic evaluation of impact of testing at different stages, but this may be underdeveloped or lack clarity.
0	0 marks No response attempted or nothing worthy of credit.

Task 5 (b)

Evaluate how well your digital products meet your audience needs, trends and preferences. Refer to your:

- completed digital products
- development process
- application of skills.

Band	AO4: Develop, test and evaluate digital products that meet the needs of specified audiences.
3	5 marks • the response is comprehensive and insightful, demonstrating an effective evaluation of how well the digital products meet audience needs, trends and preferences • the response thoroughly evaluates the completed digital products, development process and application of skills.
2	3-4 marks • the response is relevant and appropriate, demonstrating a sound evaluation of how well the digital products meet audience needs, trends and preferences • the response clearly evaluates the completed digital products, development process and application of skills, though coverage may lack some depth or detail.
1	1-2 marks • the response is basic, demonstrating a limited evaluation of how the digital products meet audience needs, trends and preferences • the response makes some reference to the completed digital products, development process and application of skills, but this may be underdeveloped or lack clarity.
0	0 marks No response attempted or nothing worthy of credit.

Appendix A: Digital Skills

Levels of complexity

*** Complex

** Medium

* Basic

Spreadsheet skills

Correcting errors:

- using built-in error checking tools *
- locating and correcting specific errors using 'Find and Replace'*
- using Data Cleaning Tools **
- adding missing data *
- use of 'Remove Duplicates' tool *
- use conditional formatting *
- apply number formatting using standard number formats *
- create and apply custom data and number formats **

Validate data against known standards or rules:

- use 'Data Validation' tool*
- create customised alert text for invalid data entries *
- create custom rules using formulas for data validation **

Sorting:

- one column using built-in sorting tools *
- sorting by multiple criteria **
- sorting based on cell or font colour **
- using formulas to create dynamic sorting ***

Categorise:

- create drop-down lists for categorisation *
- grouping and rearranging rows or columns to organise data **

Format data and tables:

- apply basic formatting such as, font styles, colour and borders *
- use built-in table styles for consistent formatting *
- use of 'Text to columns' tool *

Create and apply filters:

- using built-in AutoFilter tool*
- applying complex criteria for filtering data **
- creating custom filter criteria **
- using formulas to create dynamic filters ***

Pivot tables:

- create basic pivot tables *
- format pivot tables for better readability *
- add calculated fields and items **
- create charts from pivot table data **
- use slicers to filter pivot table data ***

Performing basic arithmetic calculations:

- addition *
- subtraction *
- multiplication *
- division *

Using basic level formulas and functions:

- SUM *
- AVERAGE *
- COUNT *
- COUNTA *
- MAX *
- MIN *
- IF *

Using medium level formulas and functions:

- Nested IFs **
- VLOOKUP **
- HLOOKUP **
- COUNTIF **
- SUMIF **
- AVERAGEIF **
- SUBTOTAL **
- SEARCH **
- FIND **

Using complex level formulas and functions:

- INDEX ***
- MATCH ***
- FILTER ***
- SORT ***
- SORTBY ***
- UNIQUE ***

Website development skills	Animation development skills	Computer game development skills
• using basic HTML (Hyper Text Markup Language) tags *	• moving shapes*	• importing sprites *
• setting up a basic webpage structure: • head * • body * • footer *	• animating simple transformations (e.g., scale, rotation, position) *	• attaching sprites to controllable objects *
• linking between pages *	• use of timeline folders *	• inclusion of a scoring system *
• creating basic navigation *	• use of timeline layers *	• creating sprites *
• including images *	• moving artwork *	• use of sounds *
• using CSS (Cascading Style Sheets) for basic styling: • fonts * • colours * • simple layouts *	• creating looping animations*	• scripting basic controls *
• including multimedia elements: • image slider ** • embedding videos ** • embedding content from external sources (e.g., YouTube videos, Google Maps) **	• applying basic visual effects (e.g., opacity changes, blurs) *	• creation of on-screen instructions *
• implementing responsive design with CSS (Cascading Style Sheets) **	• multiple different sets (locations/ backgrounds) **	• using paths for enemies **
• using flexbox and grid layouts **	• animated background **	• animating sprites **
• adding basic JavaScript: • variables and data types ** • basic control structures (e.g., if-else) ** • use of date function **	• multiple simultaneous animated events **	• creation of splash screen **

- applying basic SEO (Search Engine Optimisation) principles: - alt tags ** - meta tags ** - semantic HTML (Hyper Text Markup Language) **	lip-syncing for character dialogue**	inclusion of a health system **
- applying accessibility features: - ARIA (Accessible Rich Internet Applications) roles and attributes *** - complying with WCAG – (Web Content Accessibility Guidelines) ***	use of background sound **	- scripting advanced features: - multi player ** - attack system ** - gravity *** - attack multipliers *** - speed multipliers ***
- advanced CSS techniques: - using CSS Grid for complex layouts *** - implementing animations and transitions ***	- masking: - using masks to create reveals and hidden areas in animation ** - using multiple masks for advanced effects. ***	random object / enemy generation ***
- adding advanced JavaScript: - complex form validation*** - use of arrays *** - use of string methods *** - handling errors (e.g. try, catch, finally, throw) ***	implementing particle systems for effects. ***	creation of save states ***
	use of rigged / boned asset for animation ***	creation of a persistent leader board ***
	correctly exported as embeddable content ***	use of external controller ***
	use of subtitles to match narration ***	correctly exported as embeddable content ***
	use of narration ***	
	custom easing functions for precise control over animation timing. ***	
	interactive animation through scripting ***	

Appendix B: Mapping of questions to specification content and assessment objectives

Question	Mark allocation										
	Topic and Section								Total Marks	AO3 Marks	AO4 Marks
	2.1		2.2		2.3		2.4				
	2.1.1	2.1.2	2.2.1	2.2.2	2.3.1	2.3.2 a or b	2.4.1	2.4.2			
1 (a)	P										
1 (b)		P							5	5	0
2 (a)			P						5	5	0
2 (b)				P					5	5	0
3 (a)				P					10	10	0
3 (b)				P					5	0	5
4 (a)					P				10	0	10
4 (b)						P			25	0	25
5 (a)							P		5	0	5
5 (b)								P	5	0	5
Total marks									80	30	50